

NEBIL ESMAEL

Full-Stack Developer · AI / NLP Engineer

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PROFESSIONAL SUMMARY

Full-Stack Developer and AI/NLP Engineer with hands-on expertise in TypeScript, Node.js, NestJS, and transformer-based language-model fine-tuning. Proven track record shipping secure, scalable backend APIs and modern Next.js applications, and training multilingual models for real-world safety-classification tasks in low-resource languages. Adept at bridging deep technical work with clear documentation and mentorship.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, HTML, CSS

AI & ML: XLM-RoBERTa Fine-Tuning, LoRA/QLoRA, RAG Pipelines, HuggingFace Transformers, PyTorch, NLP Preprocessing, Dataset Curation, Safety Classification

Backend: Node.js, NestJS, Express.js, ElysiaJS, REST APIs, JWT Auth, WebSockets, Swagger/OpenAPI

Frontend: React, Next.js, Tailwind CSS, Radix/shadcn UI, Three.js, GSAP

Tools & DBs: MongoDB, Supabase, Docker, Linux, Git/GitHub, Clerk

TECHNICAL PROJECTS

Amharic Toxic Content Classifier — XLM-RoBERTa Fine-Tuning

- Fine-tuned XLM-RoBERTa on a combined HuggingFace toxic-text corpus and a custom in-house Amharic dataset, targeting child-safety classification.
- Solved low-resource NLP challenges specific to Amharic — character normalization, morphological variation, and noisy script encoding.
- Built a binary (Safe/Unsafe) classifier delivering reliable toxic-content detection for a language severely under-represented in existing safety tooling.
- Designed the full pipeline: data preprocessing and tokenization → LoRA/QLoRA training → model evaluation.

Hayyu — AI SaaS Learning Platform

[Next.js](#) · [Vapi AI](#) · [Clerk](#) · [Supabase](#)

- Built a SaaS learning platform with real-time AI voice tutors, custom subjects, a progress dashboard, and session transcripts.
- Integrated Clerk authentication, Supabase persistence, and Vapi voice sessions with Sentry monitoring.

InkLink — Social Reading & Collaborative Writing Platform

[Next.js](#) · [Yjs/TipTap](#) · [NestJS](#)

- Engineered real-time multi-user collaborative editing (Yjs/TipTap) and neural Text-to-Speech with automatic English/Amharic detection and client-side audio caching.
- Implemented wallet-based premium monetization, social profiles, and follow interactions.

Microbusiness Marketplace API

[NestJS](#) · [MongoDB](#) · [Leaflet](#)

- Designed a geolocation-aware marketplace REST API with JWT/Passport auth, request throttling, Helmet/compression hardening, and Swagger documentation.

Full-Stack Freelance Web Development

- Delivered end-to-end web applications for multiple clients with Next.js and Node.js, focused on maintainable architecture and clear documentation for non-technical stakeholders.

PROFESSIONAL EXPERIENCE

Backend Developer Intern | [Golden Age Technology PLC](#)

2025

- Designed and built scalable RESTful APIs with NestJS, integrating cleanly with modern frontend frameworks.

- Developed an internal Git repository tracking system to monitor developer commits and branch activity across the engineering team.

Backend Developer Intern | [Sematha Technology PLC](#) 2025

- Built a real-time chat application with ElysiaJS and React, managing complex WebSocket state and message-delivery logic.

Volunteer Instructor | [Jimma American Corner](#) 2024

- Taught JavaScript fundamentals and web development, making complex programming concepts accessible to beginners.
- Conducted intensive code reviews — identifying logic errors and providing written optimization feedback.

EDUCATION

B.Sc. Computer Science & Engineering · [Adama Science and Technology University](#) Expected 2026

CERTIFICATIONS

- Advanced Node.js: Level Up — Udemy (2025)
- Data Analysis Fundamentals Nanodegree — Udacity
- AI Career Essentials — ALX Ethiopia
- MongoDB Developer's Toolkit — GeeksforGeeks